

WEEK 6 PROJECT SUMMARY

Purpose. This concise summary documents the Week 6 Data Science development work, cross-track integration, key decisions, outcomes, challenges, and proposed focus for Week 7.

1. What we planned to accomplish

Move beyond the Week 5 baseline through structured false-positive/false-negative error analysis, integration of Data Analytics findings, feature refinement, evaluation of a Random Forest model, and additional validation to identify the strongest candidate model.

2. What we completed

The Week 6 workflow was completed from baseline re-establishment through final model comparison and validation. This included error analysis, cross-track integration, previous-no-show grouping, creation of a booking-lead-time × previous-no-show interaction, refined Logistic Regression, Random Forest evaluation, confusion-matrix review, and five-fold stratified cross-validation.

3. What we improved from Week 5

The Week 5 Logistic Regression baseline achieved 61.39% accuracy, 60.84% precision, 68.87% recall, 64.60% F1 score, and 0.6642 ROC-AUC. The refined Week 6 Logistic Regression achieved 61.50% accuracy, 60.87% precision, 69.28% recall, 64.80% F1 score, and 0.6647 ROC-AUC. The improvement was modest, with false negatives decreasing from 151 to 149 and true positives increasing from 334 to 336.

4. What we integrated

The main integration was the Data Analytics team's findings on booking lead time, previous no-shows, distance, appointment type, age, appointment day and time, and the interaction between booking lead time and previous no-shows. Booking lead time and previous no-shows were treated as particularly informative, while appointment time showed little variation and was not retained in the refined feature set.

5. The collaboration track(s)

The Data Science track collaborated directly with the Data Analytics track through a cross-track handoff and validation discussion, linking descriptive findings with predictive modelling and error analysis.

6. What was exchanged

Data Analytics provided subgroup-level no-show patterns, stability considerations, feature-priority guidance, and interaction findings. Data Science provided model error patterns, including false negatives and false positives across booking lead-time groups, previous-no-show levels, appointment types, age groups, days, times, and distance groups.

7. What changed as a result

The integration informed the Step 6 feature refinement stage. Previous no-shows were represented using 0, 1, 2, and 3+ groups; a booking-lead-time × previous-no-show interaction was added, and appointment time was removed because its analytical differences were essentially flat. These changes were implemented during feature refinement rather than during the handoff itself.

8. Key findings or development outcomes

Refined Logistic Regression remained stronger than Random Forest. Random Forest achieved 59.92% accuracy, 60.44% precision, 62.68% recall, 61.54% F1, and 0.6374 ROC-AUC. Five-fold stratified cross-validation also favoured refined Logistic Regression: mean ROC-AUC 0.6745 versus 0.6478 for Random Forest. The Week 6 result therefore supports retaining refined Logistic Regression as the strongest evaluated model. Greater model complexity did not automatically improve performance.

9. Major challenges

Challenges included interpreting false-positive and false-negative patterns without overstating the evidence, handling small subgroups, avoiding leakage prone variables, maintaining consistent train/test comparisons, and resolving implementation issues during cross-validation. Feature refinement also produced only modest gains, making model comparison essential.

10. Important decisions

Key decisions were to retain booking lead time and previous no-shows, group previous no-shows at 3+, add their interaction with booking lead time, remove appointment time, retain the same stratified split for direct comparison, and select refined Logistic Regression over Random Forest based on both test-set and cross-validation performance.

11. Remaining issues

The selected model has moderate predictive performance and is not deployment ready. False negatives remain important. Further work should include temporal or real-world validation, probability calibration, decision threshold optimization, segment level robustness checks, and continued monitoring. The synthetic dataset also limits generalization.

12. Contribution to the overall HealthConnect project

Week 6 transformed the Week 5 baseline into a more thoroughly evaluated predictive workflow. The work connected model errors with Data Analytics findings, refined important scheduling and behavioural features, evaluated an alternative machine learning approach, and established validation-based model selection. This provides a stronger evidence base for subsequent HealthConnect development.

13. Proposed focus for Week 7

Build on the validated Week 6 model rather than simply adding complexity. Priority areas should include deeper diagnostics, probability calibration and threshold analysis, robustness validation, targeted

investigation of remaining false negatives and false positives, and assessment of how predictions could support practical clinic interventions.

Week 6 overall outcome

Week 6 successfully progressed the HealthConnect Data Science work from a baseline model toward a validated and better understood predictive workflow. Refined Logistic Regression was selected because it consistently outperformed Random Forest on the available test and cross-validation results. The main outcome is therefore not only a small metric improvement, but a stronger evidence-based understanding of features, errors, and modelling choices that should guide the next phase.